



Tire Force and Moment Transient Response

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. This document describes the test equipment requirements, test methodology, and data format to supply tire force and moment (F&M) transient response properties required to support on-center vehicle performance simulations for the development and validation of vehicles and vehicle subsystems.

1.2 Foreword.

1.2.1 The tire F&M properties are important to various vehicle simulations groups as tire performance properties contribute greatly to the vehicle's response to specific control inputs.

1.2.2 Tire F&M properties are a key element within the tire component technical specification (CTS) and in the validation of a specific tire design to those requirements.

1.2.3 The quantification of tire F&M properties requires the measurement of the net resultant forces and moments from stresses within the tire footprint area. This implies the footprint dynamics on the measurement equipment must be similar to those experienced under actual vehicle operation. Therefore, a flat surface machine using a textured surface is required for the quantification of tire F&M properties.

1.2.4 Since the tire F&M properties are dependent upon many factors including the measurement equipment, the test surface, environmental effects, and the tire history, the test standard implementation and equipment suitability must be demonstrated prior to generating data for General Motors using this standard. The suitability demonstration will not be a formal correlation, although that may become a future requirement, but will be established through an engineering assessment of the measurement results of a tire to this standard with known performance attributes.

1.2.5 This standard involves specific sequences of mechanical solicitations of the test tire with data collection under dynamic conditions. Specific sequences of slip angle, camber angle, and spindle torque solicitation of the test tire are available and may be selected to generate information to meet specific data requirements. See Appendix B for a list of specific test sequences.

1.3 Applicability.

1.3.1 For all passenger cars and light truck tires.

1.3.2 The tire properties generated from this standard are to be determined using only new, unused, tires of the appropriate design level and conforming to all other tire CTS validation requirements unless fully discussed and agreed upon with the appropriate engineering personnel within General Motors.

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

SAE J670e

2.2 GM Standards/Specifications.

GMW15204

3 Resources

3.1 Facilities.

3.2 Equipment. The equipment must be properly maintained and calibrated to traceable standards.

3.2.1 The equipment must provide a means of loading the tire against a flat, textured surface.

3.2.2 The equipment must be capable of steering the tire a specified amount about the Z-axis (slip angle) and the X-axis (camber angle).

3.2.3 Mechanical inputs during the test include the slip angle, camber angle, spindle torque, and radial load which are to be controlled through a closed-loop feedback system. The control system must demonstrate dynamic performance sufficient to achieve the required input magnitudes and rates.

3.2.4 The roadway control system must be capable of dynamically controlling the speed, direction, and tracking of the roadway.

3.2.5 The roadway must be rigidly and evenly supported to react to the forces generated within the tire footprint. The support structure must be oriented parallel to the spin axis of the test spindle and perpendicular to the wheel plane at zero degrees of inclination angle.

3.2.6 The test surface typically consists of an abrasive material glued to a steel belt. The typical abrasive material is 120-grit 3M-ite. This product is used to simulate the texture of a typical road surface. In order to more closely represent the road surface, the surface is conditioned through stoning process and broken in prior to the initial use. See Appendix D of GMW 15204 for the details regarding new surface preparation.

3.2.7 The transducer system must measure the six forces and moments generated by the tire as well as the slip and camber angles, the loaded radius, spindle position and roadway speed as shown in Table 1.

3.2.8 The data acquisition system must acquire the output from the transducers and ultimately produce outputs presented as scaled engineering units.

3.2.9 The data acquisition method utilized for this standard will be time-based at a rate of 100 Hz, with simultaneous sampling across all channels.

3.3 Test Vehicle/Test Piece.

3.3.1 Test Specimen. Test tires must be free of all tags, tape, stones, or any other foreign material in the tread area and be in new, unused condition.

3.3.2 Test Wheel. The wheel must be selected properly to facilitate correct test results. The following should be considered when selecting the proper test wheel.

3.3.2.1 The wheel width must match that requested on the test request.

3.3.2.2 The wheel must have the proper rim flange contour for the test tire.

3.3.3 Test Surface. The roadway surface must be free of foreign material including, but not limited to, oils, rubber dust, or other deposits from previous usage.

3.3.4 Sample Size. Typical sample size is one tire.

3.4 Test Time.

Calendar time:	1 day
Test hours:	0.5 hour
Coordination hours	0.5 hour

3.5 Test Required Information.

3.5.1 The following information regarding the tire must be provided in order to facilitate the test.

- Construction number
- Tire size
- Service description
- Supplier (manufacturer)
- Load rating

3.5.2 The following information related to the desired test conditions must also be specified.

3.5.2.1 Rim width shall be the expected production wheel width.

3.5.2.2 Desired test load is to be provided if different than the standard test load described below under test conditions.

3.5.2.3 Desired inflation pressure is to be provided if different than the standard inflation pressure described below under test conditions.

Inflation pressure will be the warm operating inflation pressure for the intended application vehicle. The warm inflation pressure will typically be 20 kPa above the vehicle placard pressure for passenger car tires and 35 kPa above placard for light truck tires.

3.6 Personnel/Skills.

3.6.1 Test operators must be trained in the safe operation of the measurement equipment. The measurement equipment, by necessity, involves moving parts and loading mechanisms which must be understood and operated using proper safety precautions.

3.6.2 Operators must be trained in the selection of test components and fixtures to facilitate the proper testing of the tires.

3.6.3 Operators must be trained to observe, and correct when necessary, the test surface condition. The surface condition and contamination level directly impact the quality of the test results.

4 Procedure

4.1 Preparation.

4.1.1 Thermally stabilize the test tires by storing in an area with an ambient temperature within 5 °C of the test machine environment for at least 24 h prior to testing.

4.1.2 Mount the test tire on the test wheel.

4.1.2.1 Follow approved mounting procedures, such as those published by the Rubber Manufacturer's Association, when mounting the test tire onto test wheels.

4.1.2.2 Inspect the tire bead and the bead seat of the wheel to ensure this interface is free of damage, defects, and foreign material. Clean or replace test pieces, as necessary.

4.1.2.3 Use only a thin film of an approved mounting lubricant to facilitate proper tire mounting and bead seating. Excessive mounting lubricant can cause slippage at the tire and rim interface during testing.

4.1.2.4 Inspect the bead area of the mounted tire and wheel assembly to ensure the tire bead is firmly seated on the wheel.

4.1.3 Inflate the tire and wheel assembly 30 kPa above the required test inflation pressure and allow the tire to stabilize at room temperature.

4.1.3.1 If this is the first mounting of the test tire within the previous 24 h, then allow the assembly to stabilize for at least 3 h.

4.1.3.2 If the tire was previously mounted and inflated within the last 24 h period and is now being re-mounted, then the assembly needs to stabilize only for a period of 5 minutes which is approximately the amount of time needed to mount it onto the test machine.

4.2 Conditions.

4.2.1 Environmental Conditions. The ambient temperature in the room where the measurement equipment is operating shall be controlled to 21 ± 3 °C to minimize the thermal gradients within the test equipment and instrumentation. The room temperature gradients should also be within the equipment manufacturer recommendations which will typically be less than 1 °C/h.

4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.2.2.1 The specific test conditions for each tire to be tested to obtain the complete data set are provided in Appendix B.

4.3 Instructions.

4.3.1 Mount Tire. Mount the tire and wheel assembly onto the test equipment.

4.3.2 Set Inflation Pressure. Set the inflation pressure to the requested value. Either capped inflation pressure or regulated inflation pressure may be used.

4.3.3 Perform Break-In. Perform the break-in procedure unless the tire has been tested within the last 30 minutes. The break-in procedure is intended to relax residual, transient stresses within the tire due to packaging, storage conditions, or

other causes by exercising the structural elements of the tire.

4.3.3.1 The test sequences are provided in Appendix A for the tire break-in procedure. The table in Appendix A mimics the format of the MTS Test Definition software used on modern MTS Flat-Trac equipment.

4.3.4 Check Inflation Pressure. Check the tire and wheel assembly inflation pressure and reset to the test inflation pressure.

4.3.5 Inspect Test Surface. Inspect the test surface on the machine and remove rubber dust and debris as necessary.

4.3.6 Zero Transducers. Collect the tare readings from all measurement transducers, taking into consideration the range of slip angle and inclination angles which will be achieved during the test sequences.

4.3.7 Begin Collecting Time History Data. The data acquisition system is to be active from this point in time until Step 4.3.10 is executed.

4.3.8 Perform All Test Sequences. The test sequences are provided in Appendix B. The table in Appendix B mimics the format of the MTS Test Definition software used on modern MTS Flat-Trac equipment.

4.3.9 Stop Collecting Time History Data.

4.3.10 Test Tire Disposal. The tires used in this testing may be used for other testing as this test procedure is not considered to be destructive to the test specimen. It is common practice to follow this test procedure with GMW15204.

5 Data

5.1 Sampling and Filtering.

5.1.1 Data are to be collected and written to data files during the active sections of the testing. Data are to be reported from 4.3.8 to 4.3.10.

5.1.2 The sampling rate shall be 100 Hz using simultaneous sample and hold or bursting techniques. If the bursting technique is used, then the burst sample rate must be at least an order of magnitude higher than the base sample rate.

5.1.3 Anti-alias filtering techniques appropriate to the machine design and this standard are expected to be employed.

5.1.4 All recorded data signals are expected to utilize the same filters to eliminate phasing of signals due to filter induced phase lags.

5.1.5 Table 1 provides a list of the required data channels.

Table 1: Required Channel List

Channel Name	Description	Unit
ET	Elapsed time	seconds
SA	Lateral slip angle	degrees
IA	Inclination angle	degrees
V	Belt speed	km/h
FX	Tire longitudinal force	N
FY	Tire lateral force	N
FZ	Tire vertical force	N
MX	Tire overturning moment	N·m
MY	Spindle torque	N·m
MZ	Tire aligning moment	N·m
RL	Actual tire radius	mm
Spin Angle	Spindle angle position	Degrees
Yb	Belt position (tracking)	mm

5.2 Calculations.

5.2.1 The raw data analysis detailed below assumes the data are either acquired or has been preprocessed such that the following statements regarding the raw data, or machine generated test results, are true.

5.2.1.1 Data are represented in SI units with appropriate tare measurements subtracted and gain factors applied.

5.2.1.2 Data sign conventions are consistent with the SAE tire coordinate system as described in SAE J670e.

5.2.1.3 The data were collected using the data acquisition method described in the Sampling and Filtering section.

5.2.1.4 Data were collected while the equipment control system had control of the test and the machine response was sufficient such that the error between the command signals and the test conditions realized were small.

5.2.2 Calculations. The data processing will be described in general terms in the following paragraphs. GM will provide tire suppliers and approved third party testers with MATLAB code that can be employed to generate the necessary data reductions.

5.2.2.1 Transform the data set from the time domain into a distance domain format.

5.2.2.2 Determine the assemblies non-uniformity waveform for each of the unique test conditions of applied load and slip angles.

5.2.2.3 Remove the non-uniformity waveform from the data set.

5.2.2.4 Subtract the steady-state value prior to stopping the roadway surface from all data values within this test condition.

5.2.2.5 Normalize the data within this test condition to the steady-state value after the step steer input and the roadway has been restarted.

5.2.2.6 Fit the normalized lateral force data to a third-order transfer function shown below by employing a least-squares fitting method to minimize the total error between the measured data and a simulated step response.

$$\frac{F_Y(s)}{F_{Yss}} = \frac{1}{d_3s^3 + d_2s^2 + d_1s + 1}$$

Where:

F_Y = Tire lateral force [N]

F_{Yss} = Steady-State tire lateral force [N]

s = Distance [m]

d_1, d_2, d_3 = Transfer function denominator coefficients

5.2.2.7 Lateral Force Rise Distance. This is the distance required for the normalized lateral force response to rise from 10% to 90% of final steady-state response from the lateral force transfer function.

5.2.2.8 Fit the normalized aligning torque data to a third-order transfer function shown below by employing a least-squares fitting method to minimize the total error between the measured data and a simulated step response. The “d” coefficients in the denominator of the lateral force transfer function will be carried over to the aligning torque transfer function. Therefore, in this step, only the coefficients in the numerator need be found.

$$\frac{M_Z(s)}{M_{Zss}} = \frac{n_3s^3 + n_2s^2 + n_1s + 1}{d_3s^3 + d_2s^2 + d_1s + 1}$$

Where:

M_Z = Tire aligning torque [N·m]

M_{Zss} = Steady-State tire aligning torque [N·m]

S = Distance [m]

n_1, n_2, n_3 = Transfer function numerator coefficients

5.2.2.9 Aligning Torque Initial Value. This is the initial value of the normalized aligning torque function immediately following the step steer input, before the roadway surface begins to move.

5.2.2.10 Aligning Torque Rise Distance. This is the distance required for the normalized aligning torque response to rise from 10% to 90% of final steady-state response from the normalized aligning torque transfer function.

5.2.2.11 Aligning Torque Overshoot. This is the initial overshoot of the normalized aligning torque response transfer function. This metric will be expressed as a percent.

5.3 Interpretation of Results. Not applicable.

5.4 Test Documentation. All electronic data file formats are defined by the Global Tire-Wheel Test System (GTWTS). The file format details can be obtained from the global test coordinator for Tire-Wheel Systems or referenced from the GM Tire-Wheel Systems internal web site. All previous test data file formats are now obsolete.

When a test request is submitted in GTWTS, the tester will receive via email a copy of the Test Request in PDF form, input files and data sheets (as applicable), and XML Skeleton Data files. Electronic results files (TPC or XML formats) should be submitted for loading into the GTWTS and for generating the test report from the system. A sample report is given in Appendix D.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of the method to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

6.1 Safety Precautions.

6.1.1 This standard involves the use of test equipment with rotating components and multiple hydraulically controlled mechanisms. Operators are to be aware of those hazards and take reasonable precautions to prevent personal injury. Below are several notable areas where caution should be exercised. However, it is not intended to be an exhaustive list.

6.1.2 Make sure the roadway has come to a complete stop before approaching the machine to

change the tire and wheel assembly, spindle adapters, etc.

6.1.3 Inspect the test wheels for bent rims or excessive galling in the nut boss area.

6.1.4 Inspect fasteners for excessive wear or deformation.

6.1.5 Torque fasteners to specification when installing the tire and wheel assembly. A torque of 108.5 to 122 N·m (80 to 90 ft-lb) is recommended if not otherwise specified.

6.1.6 Do not lift large tire and wheel assemblies, such as light truck assemblies, onto the machine unassisted. A lifting device should be used for this purpose.

7 Notes

7.1 Glossary. Not applicable.

7.2 Acronyms, Abbreviations, and Symbols.

DOT	United States of America's Department of Transportation
ET	Elapsed Time (s)
F&M	Force & Moment
FX	Tire Longitudinal Force (N)
FY	Tire Lateral Force (N)
FZ	Tire Load, Vertical Force (N)
GTWTS	Global Tire-Wheel Test System
IA	Inclination Angle (degrees)
MX	Spindle Torque (N·m)
MY	Tire Overturning Moment (N·m)
MZ	Tire Aligning Torque (N·m)
SA	Slip Angle (degrees)
SAE	SAE International
T&RA	Tire and Rim Association
TIP	Test Improvement Process
TPC	Tire Performance Criteria
TWS	Tire-Wheel Systems
XML	Extensible Markup Language
Yb	Lateral Belt Tracking Position (mm)

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW16386

9 Release and Revisions

This standard was originated in June 2010. It was first approved by the Tire Global Subsystem Leadership Team in October, 2010. It was first published in November 2010.

Issue	Publication Date	Description(Organization)
1	NOV 2010	Initial publication

Appendix A: Tire Break-in Procedure

The following table is used as an end-level procedure within the MTS Flat-Trac Test Definition Software. Each line of test conditions are commanded and then the system waits for those conditions to be met within the predefined tolerance for those control channels. Once those conditions are met, the timer begins to accumulate the elapsed time. Once the elapsed time is equal to the time requirement in Column 1, the machine is the controller and then moves on to the next set of commands. This process is repeated until the list of commands is completed.

Table A1: Specific End-Level Procedure for Tire-Break-in

Time (s)	Speed (km/h)	Inclination Angle (degrees)	Slip Angle (degrees)	Load (%)	Auxiliary Command
30	7	0	0	80	Wait:Feedback
30	7	0	-1	80	Wait:Feedback
30	7	0	1	80	Wait:Feedback
30	7	0	2	80	Wait:Feedback
30	7	0	-2	80	Wait:Feedback
30	7	0	-4	80	Wait:Feedback
30	7	0	4	80	Wait:Feedback
30	7	0	0	80	Wait:Feedback
0.01	7	0	0	2	Wait:Feedback
27.58	7	0	0	160	FZRate:444
0.01	7	0	0	2	Wait:Feedback
0.01	7	0	-1	2	Wait:Feedback
27.58	7	0	-1	160	FZRate:444
0.01	7	0	-1	2	Wait:Feedback
0.01	7	0	1	2	Wait:Feedback
27.58	7	0	1	160	FZRate:444
0.01	7	0	1	2	Wait:Feedback
0.01	7	0	2	2	Wait:Feedback
27.58	7	0	2	160	FZRate:444
0.01	7	0	2	2	Wait:Feedback
0.01	7	0	-2	2	Wait:Feedback
27.58	7	0	-2	160	FZRate:444
0.01	7	0	-2	2	Wait:Feedback
0.01	7	0	-4	2	Wait:Feedback
27.58	7	0	-4	160	FZRate:444
0.01	7	0	-4	2	Wait:Feedback
0.01	7	0	4	2	Wait:Feedback
27.58	7	0	4	160	FZRate:444
0.01	7	0	4	2	Wait:Feedback
0.01	7	0	0	2	Wait:Feedback
27.58	7	0	0	160	FZRate:444
0.01	7	0	0	50	poststop

Appendix B: Test Description/Sequence

Table B1: Generalized List of Test Conditions

Step	Inclination Angle (degrees)	Speed (km/h)	Load (%)	Slip Angle (degrees)
1	0	3.6	80	+0.5
2	0	3.6	80	-0.5
3	0	3.6	80	+1.0
4	0	3.6	80	-1.0
5	0	3.6	100	+0.5
6	0	3.6	100	-0.5
7	0	3.6	100	+1.0
8	0	3.6	100	-1.0

The steps in the above table are to be run in the order presented. A series of events must be performed in a time-based sequence in order to complete each of the above steps. The test results obtained from this procedure are time sensitive, therefore, careful attention needs to be given to the timing of the events shown in Figure B1. The events denoted by the event markers can be described as follows:

- A** – Tire is operated under the load, speed, inclination angles for the test conditions and 0 degrees of slip angle.
- B** – Test speed is commanded to zero and the roadway surface is allowed to coast to a stop.
- C** – The slip angle is commanded to the appropriate test condition level.
- D** – The test speed is commanded to the appropriate test condition level to start as the slip angle is near the commanded level. Generally this is approximately 90% of the slip angle command.
- E** – Lateral force and aligning torque responses build quickly.
- F** – The slip angle is commanded back to zero degrees.

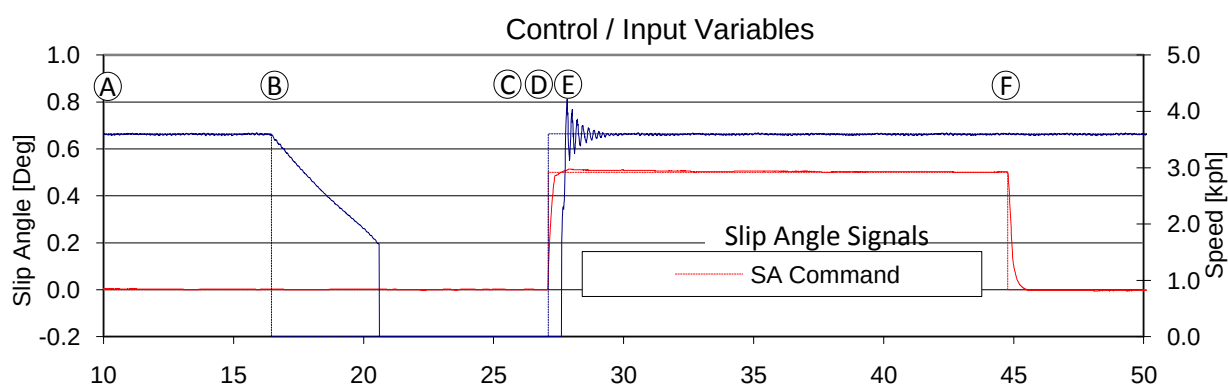


Figure B1: Time History of a Single Test Condition

Table B2: Specific End-Level Procedure Testing

Time (s)	Speed (km/h)	Inclination Angle (degrees)	Slip Angle (degrees)	Load (%)	Command	Comment
0.01	3.5	0	0.0	80	DACQON	
15	3.5	0	0.0	80	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	80	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	0.5	80		Steer Tire
15	3.5	0	0.5	80	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	80	WAIT:FEEDBACK	Get Steady-State Values
10	0	0	0.0	80	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	-0.5	80		Steer Tire
15	3.5	0	-0.5	80	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	80	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	80	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	1.0	80		Steer Tire
15	3.5	0	1.0	80	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	80	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	80	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	-1.0	80		Steer Tire
15	3.5	0	-1.0	80	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	100	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	100	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	0.5	100		Steer Tire
15	3.5	0	0.5	100	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	100	WAIT:FEEDBACK	Get Steady-State Values
10	0	0	0.0	100	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	-0.5	100		Steer Tire
15	3.5	0	-0.5	100	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	100	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	100	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	1.0	100		Steer Tire
15	3.5	0	1.0	100	WAIT:FEEDBACK	Measure Transient Response
15	3.5	0	0.0	100	WAIT:FEEDBACK	Get Steady-State Values
5	0	0	0.0	100	WAIT:FEEDBACK	Settle, Stop Belt
0.01	0	0	-1.0	100		Steer Tire
15	3.5	0	-1.0	100	WAIT:FEEDBACK	Measure Transient Response
0.01	0	0	0.0	5	~ DACQOFF	Normal exit of test

Relaxation Force & Moment Report



**Tire-Wheel
Systems**

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